Problem Sheet 5

discussion: Tuesday, 10.11.2026

5.1. Write a program that realizes the composite trapezoidal rule for integration on $[a, b]$. The rule is based on a subdivision of $[a, b]$ into N subintervals of length $h = (b - a)/N$. The inputs are a function handle for f, N, and a, b.

Consider, for $[a, b] = [-1, 1]$, the 5 integrands

$$f_1(x) = x^2, \qquad f_2(x) = |x|, \qquad f_3(x) = \begin{cases} \frac{1}{2}e^x & x < 1/3 \\ e^x & x \geq 1/3 \end{cases}, \qquad f_4(x) = \sin(\pi x), \qquad f_5(x) = \sin(4\pi x).$$

Use absolute errors and report zeros separately. Plot the non-zero errors in `log-log`-scale versus h for $h = (b - a)2^{-i}$, $i = 1, 2, \ldots, 20$. What do you observe? Explain your observations for the functions f_1, f_2, f_3.

5.2. Explain the convergence behavior in Exercise 5.1 for the integrand f_4. You may use Euler's formula $e^{\mathbf{i}z} = \cos z + \mathbf{i} \sin z$ in the form $\sin z = \frac{1}{2\mathbf{i}}(e^{\mathbf{i}z} - e^{-\mathbf{i}z})$ with the imaginary unit $\mathbf{i}$ and the geometric series

$$\sum_{j=0}^{N-1} q^j = \frac{q^N - 1}{q - 1}, \qquad q \neq 1.$$

Hint: what is the exact value of the integral?

5.3. Compute the following integral both analytically and numerically:

$$\int_{-2}^{4} 1 - x - 4x^3 + 2x^5 \, dx$$

For the numerical integration use (a) the trapezoidal rule; (b) the midpoint rule; (c) the Simpson rule; (d) the Milne rule; (e) the composite trapezoidal rule with $N = 4$; (f) the composite Simpson rule with $N = 3$.

For all methods compute the relative error $|I - Q|/|I|$ between the exact value of the integral I and the numerical approximation Q.

5.4. **a)** Show that the weights of the Newton-Cotes formulas satisfy $\sum_{i=0}^{n} w_i = 1 (=$ length of the interval $[0, 1])$.
Hint: apply the quadrature formula to a suitable function f.

b) Show the symmetry property $w_{n-i} = w_i$, $i = 0, \ldots, n$.
Hint: use the symmetry of the points, i.e., $x_j = 1 - x_{n-j}$.

c) Let $n = 2m$ be even. Consider the function $f = (x - 1/2)^{n+1}$, which is antisymmetric with respect to $1/2$. Show:

$$\int_0^1 f(x) \, dx = 0 = \widehat{Q}_n^{cNC}(f) = \widehat{Q}_n^{oNC}(f).$$

Conclude from this that the quadrature formulas $\widehat{Q}_n^{cNC}$ and $\widehat{Q}_n^{oNC}$ are exact for polynomials of degree $n + 1$!

5.5. We wish to show that the extrapolation of the composite trapezoidal rule is the composite Simpson rule. To that end, let $T(h)$ be the composite trapezoidal rule with step size $h = (b - a)/N$ and $S(h)$ be the composite Simpson rule with step size $h = (b - a)/N$. Use Romberg extrapolation with step sizes $h_i = (b - a)2^{-i}$, $i = 0, 1, \ldots$.

a) Extrapolation of the composite trapezoidal rule (with step size h) has in column $m = 0$ the values $T(h_i)$. Show: in column $m = 1$ of the Neville scheme are the values

$$N_i := T(h_{i+1}) + \frac{1}{3}(T(h_{i+1}) - T(h_i))$$

b) Show: $N_i = S(h_i)$.